

Combinatorics

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The Pigeonhole Principle

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The pigeonhole principle

- Suppose a flock(ঝাঁক) of pigeons fly into a set of pigeonholes to roost (পাখির বিশ্রামের জায়গা)
- **If there are more pigeons than pigeonholes, then there must be at least 1 pigeonhole that has more than one pigeon in it**
- If $k+1$ or more objects are placed into k boxes, then there is at least one box containing two or more of the objects
 - This is Theorem 1

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Pigeon=10

Pigeonhole=9



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Pigeonhole principle examples

- In a group of 367 people, there must be two people with the same birthday
 - As there are 366 possible birthdays
- In a group of 27 English words, at least two words must start with the same letter
 - As there are only 26 letters

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Generalized pigeonhole principle

- If N objects are placed into k boxes, then there is at least one box containing $\lceil N/k \rceil$ objects
 - This is Theorem 2

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Generalized pigeonhole principle examples

- Among 100 people, there are at least $\lceil 100/12 \rceil = 9$ born on the same month
- How many students in a class must there be to ensure that 6 students get the same grade (one of A, B, C, D, or F)?

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The “boxes” are the grades. Thus, $k = 5$
Thus, we set $\lceil N/5 \rceil = 6$
Lowest possible value for N is 30

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Rosen, section 4.2, question 4

- A bowl contains 10 red and 10 yellow balls
- a) How many balls must be selected to ensure 3 balls of the same color?
 - One solution: consider the “worst” case
 - Consider 2 balls of each color
 - You can't take another ball without hitting 3
 - Thus, the answer is 5
 - Via generalized pigeonhole principle
 - How many balls are required if there are 2 colors, and one color must have 3 balls?
 - How many pigeons are required if there are 2 pigeon holes, and one must have 3 pigeons?
 - number of boxes: $k = 2$
 - We want $\lceil N/k \rceil = 3$
 - What is the minimum N ?
 - $N = 5$

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Rosen, section 4.2, question 4

- A bowl contains 10 red and 10 yellow balls
- b) How many balls must be selected to ensure 3 yellow balls?
 - Consider the “worst” case
 - Consider 10 red balls and 2 yellow balls
 - You can't take another ball without hitting 3 yellow balls
 - Thus, the answer is 13

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Rosen, section 4.2, question 32

- 6 computers on a network are connected to at least 1 other computer
- Show there are at least two computers that have the same number of connections
- The number of boxes, k , is the number of computer connections
 - This can be 1, 2, 3, 4, or 5
- The number of pigeons, N , is the number of computers
 - That's 6
- By the generalized pigeonhole principle, at least one box must have $\lceil N/k \rceil$ objects
 - $\lceil 6/5 \rceil = 2$
 - In other words, at least two computers must have the same number of connections

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Rosen, section 4.2, question 10

- Consider 5 distinct points (x_i, y_i) with integer values, where $i = 1, 2, 3, 4, 5$
- Show that the midpoint of at least one pair of these five points also has integer coordinates
- Thus, we are looking for the midpoint of a segment from (a,b) to (c,d)
 - The midpoint is $((a+c)/2, (b+d)/2)$
- Note that the midpoint will be integers if a and c have the same parity: are either both even or both odd
 - Same for b and d
- There are four parity possibilities
 - (even, even), (even, odd), (odd, even), (odd, odd)
- Since we have 5 points, by the pigeonhole principle, there must be two points that have the same parity possibility
 - Thus, the midpoint of those two points will have integer coordinates

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